

History of PCB industry in India

- The making of PCBs in India was started by scientists and research institutes in the late 1950s.
- Bharat Electronics Limited (BEL) established the first commercial PCB manufacturing facility in India in 1968, for in-house requirements. This was followed by Indian Telephone Industries and some other defence and space research institutes.
- The first private sector firm that manufactured professional-grade PCBs came up in 1970s.
- The first boom in PCBs manufacture was between 1985 and 1990. It did not last long due to low domestic demand, lack of infrastructure, high cost of capital, low investments and lack of skilled manpower.
- The Indian Printed Circuit Board Association was set up in 1985 when a few leading PCB technocrats met in Mumbai (then known as Bombay).
- The entrance of multinational companies (MNCs) and foreign firms into the Indian market led to the second boom that started in 2001. Exports started picking up and quality reached world-class standards.

advances in technology and Indian manufacturers' inability to keep pace with them and stay ahead in R&D. This is again a function of scale and profitability.

The high cost of electricity and frequent load-shedding make it difficult to run advanced machines.

Almost the entire raw material required for PCB manufacturing is imported, out of which 90% is from China. It is believed that India's entire demand of PCBs can be met by just one major PCB plant in China. We cannot get raw materials at competitive prices due to the lack of volumes.

Since there are no manufacturers in India for copper-clad laminates, solder mask inks, photopolymer film and chemicals used in plated-through-hole (PTH) process, the manufacturing companies have to keep huge inventory of imported raw material, which blocks their working capital. Low taxes on imported PCBs has led to flooding of cheap, low-quality PCBs in the market, which is a threat to Indian manufacturers.

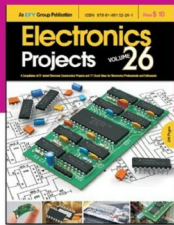
There are virtually no technical institutes in India to give training in various aspects of PCB manufacture. Hence we greatly lack the skilled manpower or specialised personnel in the field.

Foreign challenge. In India, Japanese companies like Suzuki prefer to source their PCBs from Japanese firms while Korean companies like Samsung source theirs from Korea. This results in very little business going to the Indian manufacturers. Anyway, the local players are unable to compete with Chinese exporters who get 17% export incentive.

It would be incorrect to say that the market for PCBs is not growing. The problem is that almost 75% of domestic demand for PCBs is met by imports and, in ad-

Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

It is a compilation of 21 construction projects and 71 circuit ideas published in Electronics For You magazine.



This collection of tested circuit ideas and construction projects in a handy volume would interest all classes of electronics enthusiasts—be they students, teachers, hobbyists or professionals!

Electronics Projects Vol. 26 is available on the following digital magazine stores:

- www.lulu.com
- www.magzter.com
- pocketmags.com
- www.getscop.com
- rockstand.in
- www.studybuddy.com
- www.readwhere.com
- pothi.com